

04 · Which funnel stage to invest in? — mediation (pathmc)

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The decision. A fixed budget to split across the funnel — **onboarding -> engagement -> activation -> conversion**. Which stage gives the most leverage on paid conversion, *respecting that effects cascade*? A dollar into onboarding doesn't just help onboarding; it flows downstream. We need to trace those paths, decompose **direct vs indirect** effect, and — crucially — know **how fragile the split is** to unmeasured confounding between a mediator and the outcome (the classic mediation trap).

pathmc builds the chain as a structural causal model and computes **path-specific effects** by propagating posterior draws through the intervened graph — the modern, uncertainty-aware mediation analysis. 7-step contract.

1.1 2 · Simulate a ground truth

A chain `onboarding_score -> engagement -> activated -> converted`, with `channel_quality` feeding engagement and activation. The planted structure gives a known **total** effect of onboarding on conversion, split into paths — most flows through `engagement -> activated`, a little through `engagement -> converted` directly.

TRUE effects of onboarding on conversion:

<code>indirect_via_engagement_activated</code>	3.17
<code>indirect_via_engagement_only</code>	0.77
<code>total</code>	3.94

	<code>onboarding_score</code>	<code>channel_quality</code>	<code>engagement</code>	<code>activated</code>	<code>converted</code>
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

1.2 3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \rightarrow M \rightarrow Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$\text{TE} = \text{NDE} + \text{NIE}, \quad \text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a linear SCM these are the **product-of-coefficients**: indirect = $a \cdot b$, direct = c . The **proportion mediated** = NIE/TE tells you how much of onboarding’s value is “really” downstream engagement/activation. **pathmc** computes path-specific effects for the full chain by simulating through the intervened graph.

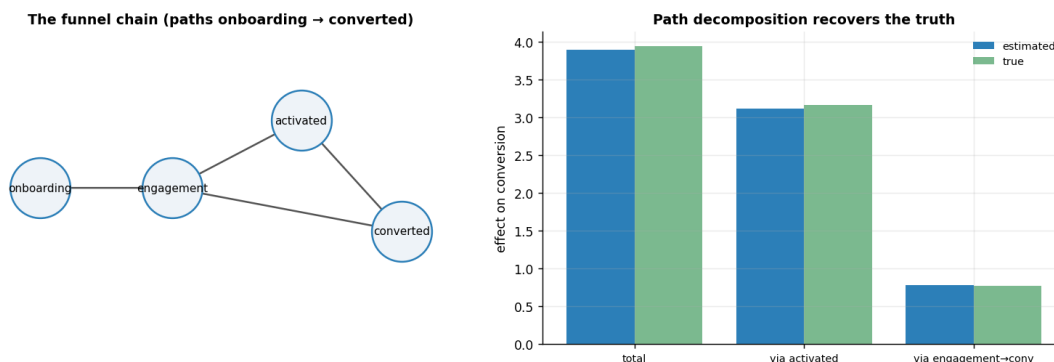
Assumptions (stronger than a single-link analysis): no unmeasured confounding of $X-Y$, $M-Y$, and $X-M$, plus the **cross-world** condition that no $M-Y$ confounder is itself affected by X . Natural effects are *not identified* if that last one fails — and the $M-Y$ confounding assumption is the one most likely to break, so we stress-test it in step 6.

1.3 4 · Estimate — fit the chain and query every path

	mean	sd	hdi_3%	hdi_97%
name				
e_on	2.150	0.042	2.073	2.226
e_ch	1.131	0.043	1.043	1.207
a_eng	1.616	0.018	1.584	1.652
a_ch	0.592	0.054	0.490	0.690
c_act	0.896	0.022	0.857	0.940
c_eng	0.361	0.040	0.286	0.430

1.4 5 · Validate — recover the total, the paths, and the proportion mediated

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TOTAL onboarding -> converted          3.89 (true 3.94) [90% 3.74, 4.03]
  via engagement->activated->conv      3.11 (true 3.17) [90% 2.95, 3.28]
  via engagement->converted             0.78 (true 0.77) [90% 0.63, 0.92]
Proportion mediated (indirect/total): 100% [99%, 101%] - most of onboarding's
value is realised downstream.
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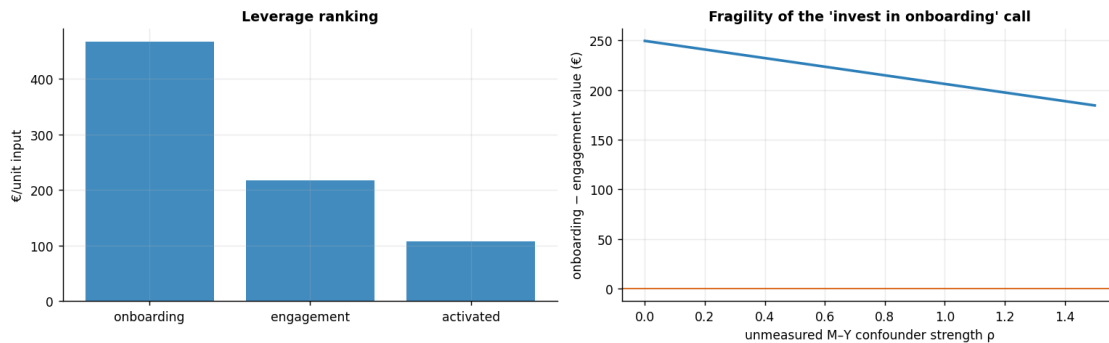
1.5 6 · Decide, in euros — leverage ranking *and* its fragility

Two moves. First, **rank levers** by the € of conversion value from a unit improvement in each stage’s input (using the total downstream effect — onboarding’s leverage cascades). Second, the honest part: **sensitivity of the mediation split to unmeasured $M-Y$ confounding**. If an omitted trait drives both engagement and conversion, the indirect path is over-credited — we sweep

a confounding strength and show how far the “invest in onboarding vs engagement” call would have to be pushed before it flips.

lever	€/unit	P(>0)
onboarding	467.0	1.0
engagement	217.0	1.0
activated	107.0	1.0

Onboarding leads on leverage, and the call is ROBUST: even the strongest M-Y confounding we swept ($\rho=1.5$) doesn't close the €250 gap to engagement.



1.6 7 · Caveats

- **Natural effects lean on strong assumptions** — no unmeasured confounding of any of the three links plus the cross-world condition. Step 6 quantified how much $M \rightarrow Y$ confounding would flip the decision; the point estimate alone hides that fragility.
- **Don't control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you're pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear.** For strong nonlinearities lean on `pathmc`'s simulate-through-the-graph numbers, not hand-multiplied coefficients.